

zkEVM Prover Infrastructure

Hardware Procurement Strategy: Build vs. Buy vs. Rent

32 GPUs · 8 Servers · 2 Proving Clusters

February 2026



Executive Summary



DIY Build
(Recommended)

~€247K

CapEx for 8 servers



Prebuilt
Servers

€296-376K

CapEx for 8 servers



Cloud
Rental

€240K/yr

32 GPUs @ €20K/mo



Colo
OpEx

~€120K/yr

Setup + hosting



Recommendation

Self-build 8 servers with RTX 5090 GPUs

3-year TCO of ~€570K (DIY + colo) vs. ~€680K for cloud rental — saving ~€110K over 3 years while owning the hardware.

First-year cost: ~€367K (CapEx) + ~€120K (colo) = ~€487K.

System Architecture

2 Proving Clusters × 16 GPUs = 32 GPUs Total

Cluster A — 16 GPUs



Server 1

1× 9975WX · 4× RTX 5090 · 256GB



Server 2

1× 9975WX · 4× RTX 5090 · 256GB



Server 3

1× 9975WX · 4× RTX 5090 · 256GB



Server 4

1× 9975WX · 4× RTX 5090 · 256GB

Cluster B — 16 GPUs



Server 5

1× 9975WX · 4× RTX 5090 · 256GB



Server 6

1× 9975WX · 4× RTX 5090 · 256GB



Server 7

1× 9975WX · 4× RTX 5090 · 256GB



Server 8

1× 9975WX · 4× RTX 5090 · 256GB

NUMA constraint: 1 CPU can optimally connect to max 4 GPUs — requires min. 8 high-frequency cores per GPU (32 cores for 4 GPUs)

Non-Negotiable Requirements

✓ 19" rack mountable, 4U form factor

✓ Front-to-back airflow

✓ Dual PSU (each able to run full system)

✓ Dual 10G SFP+ NIC (or 100G QSFP28)

✓ Dedicated IPMI/BMC management port

✓ 2× USB-A, VGA display output

✓ Orderable from German entity (GmbH)

✓ Shipping to Germany

✓ Min. 8 CPU cores per GPU (high frequency)

✓ DDR5 ECC registered memory

Option 1: DIY Build — Per Server Specification

RECOMMENDED

Component	Specification	Est. Price (EUR)
GPU	4× ASUS TUF RTX 5090 OC (32GB GDDR7, 3.5 slots)	€14,800
CPU	AMD Threadripper PRO 9975WX (32C/64T, 4GHz base, 5.4GHz boost)	€3,800
Motherboard	ASRock WRX90 WS EVO (7× PCIe 5.0 x16, 8 DIMM, SSI EEB)	€900
RAM	8× 32GB DDR5 6400 ECC RDIMM (256GB total)	€8,000
Storage	2× Samsung 9100 PRO 2TB NVMe (RAID 1)	€640
NIC	Mellanox ConnectX-5 / HPE 841QSFP28 (2× 100G QSFP28)	€400
Chassis + PSU	OCG4780-4H5-H (4U, dual 3200W PSU, ATX/eATX/EEB)	€2,300
Total per Server		~€30,840
Total 8 Servers	32× RTX 5090 GPUs across 2 clusters	~€246,720



Note: Chinese chassis vendor (OneChassis) — various lead times, compatibility not fully verified. Ordering may require extra coordination.

Option 2: Prebuilt Server Quotes

	Supermicro BTO	Thomas Krenn	Primeline (Gigabyte)	Supermicro EU
GPU	4× RTX PRO 6000 Max-Q (300W)	4× RTX PRO 6000 (600W)	4× RTX PRO 6000 Max-Q	4× RTX PRO 6000 (configurable)
CPU	TR PRO 9945WX 12C 4.7GHz	EPYC 9374F 32C 3.85GHz	2× EPYC 9124 16C 3.0GHz	2× EPYC 9124 16C
RAM	64GB DDR5	64GB DDR5 ECC Reg	96GB DDR5 ECC Reg	128GB DDR5
Storage	2× 480GB NVMe	2× 1.92TB NVMe	4× 1.92TB NVMe	4× 1.9TB U.2 NVMe
NIC	100GbE 2× QSFP28	100GbE Dual QSFP56	2× 10GbE	Included
PSU	2600W Redundant	2× 3000W Titanium	Included	4× 2000W
Price/Server	€37,000	€47,100	€40,598	€41,559
Price × 8	€296,000	€376,800	€324,786	€332,471

Key differences: Supermicro BTO only offers Max-Q (300W) GPUs. Thomas Krenn has full 600W RTX PRO 6000 but is most expensive. Primeline uses dual-CPU Gigabyte chassis but only 10GbE networking. All prebuilts use RTX PRO 6000 (96GB) vs. RTX 5090 (32GB) in DIY.

Option 3: Cloud GPU Rental

Rental Pricing

8 GPUs = €5,000 / month

32 GPUs = €20,000 / month

Annual cost: €240,000

Max discount: 5.5% (1-year commitment)

Discounted annual: ~€226,800

Advantages

- No upfront CapEx
- No colo costs or hardware management
- Flexible scaling up/down

Disadvantages

- Most expensive over 3 years (~€720K)
- No asset ownership / residual value
- Dependency on vendor availability & pricing

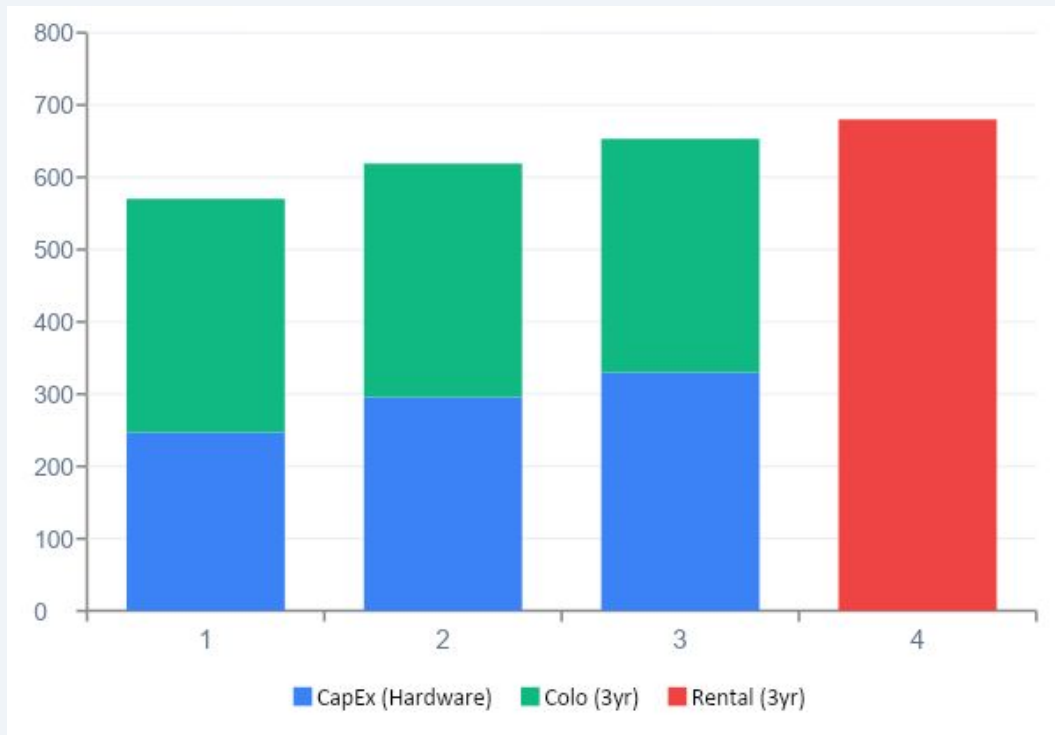
3-Year Rental Cost

Without discount: $€240K \times 3 = €720,000$

With 5.5% discount (1yr commitment): $\sim €226.8K \times 3 = €680,400$

No hardware asset at end of term. Price subject to vendor changes.

3-Year Total Cost of Ownership



DIY Build

~€570K

Saves ~€110K

€247K HW + €323K Colo

Prebuilt (Cheapest)

~€619K

€296K HW + €323K Colo

Prebuilt (Avg)

~€653K

€330K HW + €323K Colo

Cloud Rental

~€680K

Most expensive

€226.8K/yr × 3 (w/ discount)

Colo: €16.5K setup + €8.5K/mo × 36mo = €322.5K. Rounded to €323K. Rental assumes 5.5% annual discount.

Colocation Hosting



One-Time Setup

€16,500

Custom power wiring to racks required



Monthly Operating

~€8,500 / mo

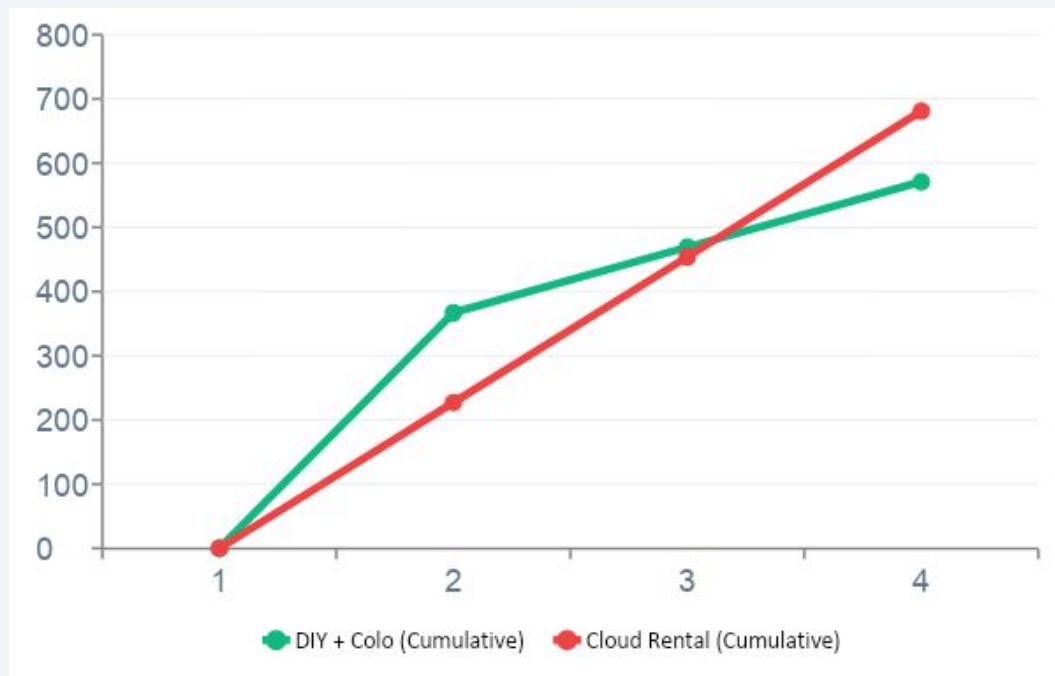
Racks, power, and cooling included

Annual Breakdown

	Setup	Monthly × 12	Annual Total
Year 1	€16,500	€102,000	€118,500 (~€120K)
Year 2	—	€102,000	€102,000
Year 3	—	€102,000	€102,000
3-Year Total			€322,500 (~€323K)

3-year commitment required

DIY Build: Year-by-Year Cost Projection



Year 1

DIY: ~€367K

Rental: ~€227K

CapEx €247K + Colo €120K

Year 2

DIY: ~€102K

Rental: ~€227K

Colo only

Year 3

DIY: ~€102K

Rental: ~€227K

Colo only

3-Year Total

DIY: ~€571K

Rental: ~€681K

DIY saves ~€110K



Breakeven: ~22 months. After Year 1, DIY costs only ~€102K/year (colo) vs. ~€227K/year (rental). Hardware becomes a depreciable asset with potential residual/resale value. Year 1 is front-loaded with CapEx but annual OpEx is 55% less than rental.

Recommendation & Next Steps



Proceed with DIY Build + Colocation

8 servers × 4 RTX 5090 GPUs, Threadripper PRO 9975WX, 256GB DDR5 ECC per server

Total CapEx: ~€247K | Year 1 OpEx: ~€120K | 3-Year TCO: ~€571K

Saves ~€110K vs. cloud rental over 3 years while retaining hardware ownership

Next Steps

- 1 Approve budget: ~€487K Year 1 (€247K hardware + €120K colo + contingency)
- 2 Sign colocation contract (3-year commitment) & begin custom power wiring
- 3 Order hardware: source chassis, GPUs, CPUs, and components from German suppliers
- 4 Build, test, and deploy — verify GPU/CPU/NUMA compatibility before rack install

Appendix: CPU Selection Rationale

	TR PRO 9975WX (Selected)	TR PRO 9985WX	TR 9970X (Original Doc)
Cores / Threads	32 / 64	64 / 128	32 / 64
Base Clock	4.0 GHz	3.2 GHz	4.0 GHz
Boost Clock	5.4 GHz	5.4 GHz	5.1 GHz
L3 Cache	160 MB	256 MB	64 MB
Memory	8-ch DDR5 6400	8-ch DDR5 6400	4-ch DDR5 5600
PCIe Lanes	128 PCIe 5.0	128 PCIe 5.0	48 PCIe 5.0
TDP	350W	350W	350W
Cores per GPU	8 (perfect fit)	16 (overkill)	8 (tight fit)
Socket	sTR5 (WRX90)	sTR5 (WRX90)	sTR5 (TRX50)
Price (EUR)	~€3,800	~€7,500	~€2,550

Why 9975WX: Provides exactly 32 cores (8 per GPU) with the highest base clock (4.0 GHz) in the lineup. The 9985WX doubles cores but drops to 3.2 GHz base — unnecessary for 4 GPUs and nearly 2× the price. The original 9970X lacks WRX90 PRO platform features (8-channel memory, 128 PCIe lanes).